WORKSHEET 0.3

Exponentials, Logarithms and Their Laws

Unit 0: Readiness

Key ideas

- $y = \ln x$ means $e^y = x$. So $\ln(e^x) = x$ for all x, and $e^{\ln x} = x$ for $x > 0$.
- $\ln(ab) = \ln a + \ln b$, $\quad \ln\dfrac{a}{b} = \ln a - \ln b$, $\quad \ln\left(a^b\right) = b\ln a$. There is no law for $\ln(a+b)$.
- To solve for an exponent: **isolate** the exponential, then take the natural logarithm.
- A logarithmic equation can produce a value that is not allowed. **Check** every solution in the original equation.

Worked examples

Example 1 · Simplifying

Simplify $e^{\ln 4 + \ln 3}$ and $\ln\left(e^2\sqrt{e}\right)$.

Solution. $e^{\ln 4+\ln 3} = e^{\ln 12}$. $\ln\left(e^2 \cdot e^{1/2}\right) = \ln\left(e^{5/2}\right)$.

12 and $\dfrac{5}{2}$.

Example 2 · Isolate, then take logs

Solve $5e^{2x} - 3 = 17$.

Solution. $5e^{2x} = 20$, so $e^{2x} = 4$ and $2x = \ln 4$.

$x = \dfrac{\ln 4}{2} = \ln 2$.

Example 3 · A difference of logarithms

Solve $\ln(x+6) - \ln x = \ln 4$.

Solution. $\ln\dfrac{x+6}{x} = \ln 4$, so $x + 6 = 4x$.

$x = 2$, which makes both logarithms defined.

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Practice

CORE

1. Simplify $\ln(e^7) - e^{\ln 2}$.

2. Solve $e^x = \dfrac{1}{2}$.

3. Solve $\ln x = -1$.

4. Write $2\ln a - \ln b + \ln c$ as a single logarithm.

EXAM LEVEL

5. Solve $e^{2x} - 9e^x + 20 = 0$.

6. Solve $\ln(x-2) + \ln(x+2) = \ln 5$.

7. Solve $3^x = 15$, giving an exact answer in terms of natural logarithms.

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8. Write $\ln \dfrac{x^3 e^x}{\sqrt{x^2+1}}$ as a sum and difference of simpler terms.

Question 9 · Free-response practice

The amount of a drug in a patient's bloodstream t hours after an injection is $A(t) = 80e^{-0.1t}$ milligrams.

(a) How much of the drug is present at $t = 0$?

(b) Find the exact time at which 20 milligrams remain.

(c) Find the exact half-life of the drug, and explain how it is related to your answer to part (b).

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Worked solutions

1. $7 - 2 = 5$.
2. $x = \ln \dfrac{1}{2} = -\ln 2$.
3. $x = e^{-1} = \dfrac{1}{e}$.
4. $\ln \dfrac{a^2 c}{b}$.
5. $(e^x - 4)(e^x - 5) = 0$, so $x = \ln 4$ or $x = \ln 5$.
6. $\ln(x^2 - 4) = \ln 5$ gives $x^2 = 9$. Only $x = 3$ is allowed, since $x = -3$ makes the logarithms undefined.
7. $x \ln 3 = \ln 15$, so $x = \dfrac{\ln 15}{\ln 3}$.
8. $3 \ln x + x - \dfrac{1}{2} \ln(x^2 + 1)$.
9. (a) $A(0) = 80$ milligrams.

   (b) $80e^{-0.1t} = 20$ gives $e^{-0.1t} = \dfrac{1}{4}$, so $t = 10 \ln 4$ hours (about 13.9).

   (c) $e^{-0.1t} = \dfrac{1}{2}$ gives $t = 10 \ln 2$ hours. Falling from 80 to 20 is two halvings, and $10 \ln 4 = 2 \cdot 10 \ln 2$.

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